nodal or elemental shear moduli. If the shear modulus is nodally defined, we can use the same interpolation function with the displacement field. In this case, the shear modulus can be expressed as:

$$\mu \approx \mu^h = \sum_{i=1}^{N} N_i \, \mu_i \tag{18}$$

where N is the total number of nodes in the finite element model and N_i is the finite element shape function. μ_i is the nodal shear modulus.

In the explicit inverse method, since shear modulus distribution is a function of geometric parameters and the shear modulus in each region, the gradient of the objective function with respect to the optimization variables can be calculated using the chain rule. To this end, we should seek $\delta\mu$ corresponding to a change in any optimization variable from β to $\beta + \delta\beta$:

$$\delta\mu = \frac{\partial \mu}{\partial \beta} \, \delta\beta \tag{19}$$

Thereby, comparing to the implicit inverse method, we need to evaluate the $\partial\mu/\partial\beta$.

(a) Layered structure

The evaluation of the gradient of geometric optimization variables in a layered structure with $n+1$ layers should be discussed considering three cases. For the m-th geometric optimization variable r_m^1 at the first interface C^1, $\frac{\partial \mu}{\partial r_m^1}$ is given by:

$$\frac{\partial \mu}{\partial r_m^1} = \left[-\mu^0 + \mu^1 \big(1 - H\big(\phi^2(\mathbf{x})\big)\big) \right] \frac{\partial H\big(\phi^1(\mathbf{x})\big)}{\partial \phi^1(\mathbf{x})} \, \frac{\partial \phi^1(\mathbf{x})}{\partial r_m^1} \tag{20}$$

For the m-th geometric optimization variable r_m^n at the last interface C^n, $\frac{\partial \mu}{\partial r_m^n}$ is given by:

$$\frac{\partial \mu}{\partial r_m^n} = \left(-\mu^{n-1} \, H\big(\phi^{n-1}(\mathbf{x})\big) + \mu^n \right) \frac{\partial H\big(\phi^n(\mathbf{x})\big)}{\partial \phi^n(\mathbf{x})} \, \frac{\partial \phi^n(\mathbf{x})}{\partial r_m^n} \tag{21}$$

For the m-th geometric optimization variable at the intermediate interface $C^i \, (2 \leq i \leq (n-1))$, $\frac{\partial \mu}{\partial r_m^i}$ is given by:

$$\frac{\partial \mu}{\partial r_m^i} = \left[-\mu^{i-1} \, H\big(\phi^{i-1}(\mathbf{x})\big) + \mu^i \big(1 - H\big(\phi^{i+1}(\mathbf{x})\big)\big) \right] \frac{\partial H\big(\phi^i(\mathbf{x})\big)}{\partial \phi^i(\mathbf{x})} \, \frac{\partial \phi^i(\mathbf{x})}{\partial r_m^i} \tag{22}$$

In Eqs. (20–22), $\frac{\partial H(\phi^j(\mathbf{x}))}{\partial \phi^j(\mathbf{x})}$ $(1 \leq j \leq n)$ can be evaluated by Eq. (13) as the Heaviside function is approximated by a continuous and differentiable function. $\frac{\partial \phi^j(\mathbf{x})}{\partial r_m^j}$ can be evaluated by the finite difference method.

For the shear modulus of the background μ^0, $\frac{\partial \mu}{\partial \mu^0}$ is given by:

$$\frac{\partial \mu}{\partial \mu^0} = 1 - H\big(\phi^1(\mathbf{x})\big) \tag{23}$$

For the shear modulus of the last layer μ^n, we have

$$\frac{\partial \mu}{\partial \mu^n} = H(\phi^n(\mathbf{x})) \tag{24}$$

For the shear modulus of the layer μ^i $(1 \leq i \leq (n-1))$, we have

$$\frac{\partial \mu}{\partial \mu^i} = H\big(\phi^i(\mathbf{x})\big)\left(1 - H\big(\phi^{i+1}(\mathbf{x})\big)\right) \tag{25}$$

(b) Domain with inclusion embedded

For each m-th geometric optimization variables of j-th component r_m^j, $\frac{\partial \mu}{\partial r_m^j}$ is given by:

$$\frac{\partial \mu}{\partial r_m^j} = \left[\mu^0 \prod_{\substack{i=1 \\ i \neq j}}^{n} H\big(\phi^i(\mathbf{x})\big) - \mu^j\right] \frac{\partial H\big(\phi^j(\mathbf{x})\big)}{\partial \phi^j(\mathbf{x})} \frac{\partial \phi^j(\mathbf{x})}{\partial r_m^j} \tag{26}$$

For the shear modulus of the background μ^0, we have: